SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I T. Pradeep (192...47) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T. Pradeep.

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	1	1	1
Application	20%	1	1	1	1
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	1	1	1	1

Total Marks 29 /40

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of

3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organizatio n	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professiona l presentatio n	Understanda ble with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

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Computer Networks

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1.) A Company need to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwith of 50Mbps. The Transport layer divides the file into 1500-byte segment while the Data link layer encapsulates each segment into frames by adding 40 byte of header and trailer. Calculate the total number of segments and frames transmitted. Explain how we the transport and Data link layer ensures reliable communication.

2.) A Transport layer Protocal uses a sliding window size of 6 frames each frame carries 1024 bytes to user data. If the sender transmities 48 frames. Calculate the number of transmission windows required. If one frame in every window. Explain how Slow Control improves communication effeciency.

3.) A 12000-byte IP Packet is transmitted through a router where the max transmission unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-bytes IP header. Calculate the total number of fragments created and the total headed overhead introduced. Explain how the network layer ensures

Successfully delivery of corrected Packets to de

4) A Transport layer Protocal uses a sliding window size of 6 frames each frame carries 1024 bytes of user data. If the sender transmits 48 frames, Caclulate the number of transmissionewindows reeu If one frame in every window is lost and must be retransmitted. determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Q1) File transfer - 150MB, Bw 50MBPS, segment 1500MB DL overhead 40B

Step 1:- segment and Frames

Formula:-

NO. of segments = File size / MSS

File = 150MB = 150×10^6B = 150,00,000

Segments = 150,000,000 / 1500 = 100,000 seu

If using binary 1MB = 1024^2:

$150 \times 1024 \times 1024$ = 157,286,400B / 1500 =

Each Transport Segment = 1 Data link frame

So, Farames = 100,000 (or 1,04,858)

Step2:- Data link overhead

overhead = Frames × 40 = 100,000 × 40

Binary: 1,04,858 × 40 = 4.19 MB.

Step3:- Transmission Time

Total data on wire = File + overhead = 150+4 = 154MB

In bit = 154MB × 8 = 1232 Mb

Tt = Total Bits/Bandwidth = 1232/50 = 24sec

Without overhead:-

150MB = 1200Mb/50 = 24sec

overhead add 0.64sec

Step4:- Reliability

*Transport layer: connected oriented TCP. uses sequence number to order segment numbers to order segments, Ack and Retransmission Timer to recover lost segments, Checksum to detect errors, sliding windows for flow control.

Data link layer:- uses Framing to define the boundaries, CRC/FCS in trailer to detect bit errors, MAC Ack and CSMA/CD. Ensures error-free hop-to-hop delivery

8d) Sliding window size 6, 1024B Per frames, 48 frames.

Step1:- windows required

No.of windows = Total Frames/windows

each window transmission takes 1RTT. So 8RTT needed

Step2:- Retransmissions

Given 1 frame lost Per window.

Retransmission = 1 X No.of windows

Total frames actually sent on medium = 48+8 = 56 frames.

Total user data = 48 X 1024 = 49,152B.

Efficiency loss = 8/48 = 16.66%

Step3:- Flow control Importance

Flow control Prevents fast sender from overwhelming slow receiver. Sliding window allows sender to send w frames without waiting. Receiver advertises its buffer size If buffer full, window size = 0, sender stops. This avoids Packets drop, reduces retransmittion Prevents Congestion.

12000B IP Packet, MTU 1500B

step1:- Fragmentation

MTU includes IP header. header = 20B.

Payload Capacity Per fragment = 1500 - 20 = 1480B

Orginal Data = 12000 - 20 = 11980B

No. of fragments = Ceil(11980/1480) = Ceil

8 fragments carry 1480B data, last carries 140B

Each fragment Size : 8 × 1500B, 1 × 160B (140+20)

step 2:- overhead

Total header sent = 9 × 20 = 180B

Orginal header = 20B

Extra overhead introduction = 160B

step3:- Reassembly

Network layer uses 3 fields: Identification same ID for fragments, MF bit - 1 for 8, 0 for last, fragment Offset - offset/8 bytes to order fragment. Destination uses ID + source IP to group, orders by offset reassembles.

Sliding window size 6, 1024 B Per frames, 48 frames

Step1:- windows required

No.of windows = total frames / windows

Each window transmission takes 1 RTT. So 8 RTT needed

Step2:- Retransmissions

Given 1 frame lost Per window

Retransmission = 1 × No.of windows.

Total frames

Session layer checkPointing

Given: File = 600MB, checkPoint every 60 MB, failure at 390MB.

* checkPoints before failure:

floor(390/60) = 6 checkPoint at 60, 120, 180, 2

* Retransmission after recovery:- only from last checkPoint:- 390 − 360 = 30MB needs to be retransmitted.

without checkpoints, 390MB would be retran

Saving = 360MB.